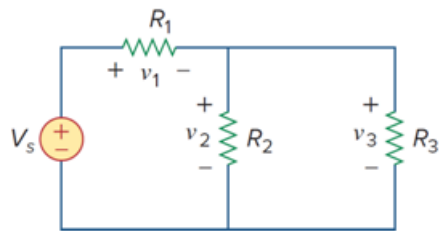


Problem

Design a problem, using Fig. 2.92, to help other students better understand series and parallel circuits.

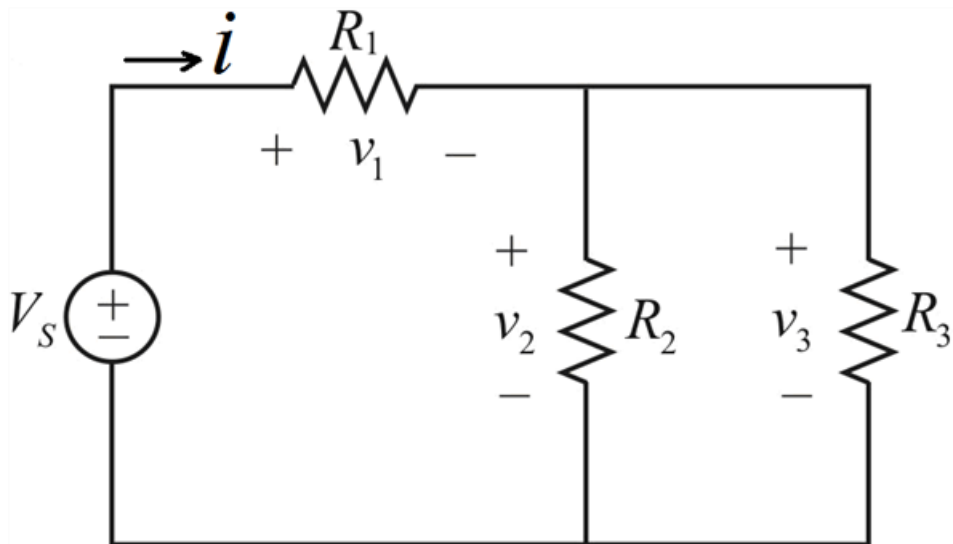
Figure 2.92:



Step-by-step solution

Step 1 of 6

The given circuit is



Step 2 of 6

R_2 resistor is parallel to R_3 resistor, so their combined resistance is

$$R_2 \parallel R_3 = \frac{R_2 R_3}{R_2 + R_3}$$

R_1 is in series with $(R_2 \parallel R_3)$, so their equivalent resistance is

$$R_{\text{eq}} = \frac{R_2 R_3}{R_2 + R_3} + R_1$$

$$R_{\text{eq}} = \frac{R_2 R_3 + R_1 R_2 + R_1 R_3}{R_2 + R_3}$$

Step 3 of 6

From Ohm's law,

$$V_s = iR_{\text{eq}}$$

$$i = \frac{V_s}{R_{\text{eq}}}$$

$$i = \frac{V_s}{\left(\frac{R_2 R_3 + R_1 R_2 + R_1 R_3}{R_2 + R_3} \right)}$$

$$i = \frac{V_s (R_2 + R_3)}{R_2 R_3 + R_1 R_2 + R_1 R_3}$$

Step 4 of 6

Now, $v_1 = iR_1$

$$v_1 = \frac{V_s (R_2 + R_3)}{R_2 R_3 + R_1 R_2 + R_1 R_3} \times R_1$$

$$\therefore v_1 = \frac{R_1 (R_2 + R_3)}{R_2 R_3 + R_1 R_2 + R_1 R_3} V_s$$

Step 5 of 6

Applying KVL to the left loop, we obtain

$$v_1 + v_2 - V_s = 0$$

$$v_2 = V_s - v_1$$

$$v_2 = V_s - \frac{(R_2 + R_3) R_1}{R_2 R_3 + R_1 R_2 + R_1 R_3} V_s$$

$$v_2 = V_s \left[1 - \frac{R_1 R_2 + R_1 R_3}{R_2 R_3 + R_1 R_2 + R_1 R_3} \right]$$

Step 6 of 6

$$\therefore v_2 = \left[\frac{R_2 R_3}{R_2 R_3 + R_1 R_2 + R_1 R_3} \right] V_s$$

Since resistors R_2 and R_3 are in parallel, voltage across them is equal i.e., $v_2 = v_3$.

$$\therefore v_3 = \left[\frac{R_2 R_3}{R_2 R_3 + R_1 R_2 + R_1 R_3} \right] V_s$$